Module 2
Management Accounting

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

Calculation of equivalent units:

	Total	Materials transferred-in	Materials added	Conversion costs
Completed units (W1)	219,000	219,000	219,000	219,000
Closing WIP	20,000	20,000	16,000	8,000
Opening WIP	(50,000)	(50,000)	(35,000)	(30,000)
Normal loss (W2)	10,000	0	0	0
Abnormal loss (W3)	1,000	1,000	1,000	1,000
Equivalent units	200,000	190,000	201,000	198,000

Workings

(W1) (50,000 + 200,000 – 11,000 – 20,000) units = 219,000 units
(W2) (200,000 × 5%) units = 10,000 units
(W3) (11,000 – 10,000) units = 1,000 units

Answer 1(b)

Calculation of cost per equivalent unit:

	Materials transferred-in	Materials added	Conversion costs
Costs incurred in last month	HK$380,000	HK$186,930	HK$112,860
Equivalent units	190,000	201,000	198,000
Cost per equivalent unit	HK$2	HK$0.93	HK$0.57

Answer 1(c)

Cost of units transferred to next process:

	HK$
Opening WIP	185,000
Costs incurred to complete the opening WIP:	
Materials added (50,000 × 30% × HK$0.93)	13,950
Conversion costs (50,000 × 40% × HK$0.57)	11,400
Units started and completed in current period [(219,000 – 50,000) × HK$(2 + 0.93 + 0.57)]	591,500
	801,850

Cost of abnormal loss:

1,000 × HK$(2 + 0.93 + 0.57)	3,500

Cost of closing WIP:

Materials transferred-in (20,000 × 100% × HK$2)	40,000
Materials added (20,000 × 80% × HK$0.93)	14,880
Conversion costs (20,000 × 40% × HK$0.57)	4,560
	59,440

Answer 1(d)

Joint products are two or more products which are separated from a common processing operation, each having a significant saleable value to warrant recognition as a main product.

The other products of the same common processing operation that have comparatively low sales value are called by-products.

Answer 2(a)

Budgeted overheads are absorbed on a machine hour basis:

Total machine hours in next year:

	Hours
Premium	200,000
Deluxe	300,000
	500,000

Production overhead absorption rate:
HK$6,400,000 ÷ 500,000 = HK$12.80 per machine hour

Selling price per unit:

	Premium	Deluxe
	HK$	HK$
Direct materials	20.00	10.00
Direct labour	30.00	60.00
Prime costs	50.00	70.00
Production overheads	25.60	19.20
Production costs	75.60	89.20
Mark-up (25%)	18.90	22.30
Selling price	94.50	111.50

Answer 2(b)

Cost driver rates:

	HK$
Machinery overheads per machine hour (HK$2,000,000 ÷ 500,000)	4
Set up overheads per set up [HK$1,800,000 ÷ (410 + 190)]	3,000
Materials handling overheads per requisition [HK$1,400,000 ÷ (920 + 1,080)]	700
Inspection overheads per inspection [HK$1,200,000 ÷ (670 + 330)]	1,200

Selling price per unit:

	Premium	Deluxe
	HK$	HK$
Prime costs	5,000,000	14,000,000
Production overheads:		
Machinery	800,000	1,200,000
Set up	1,230,000	570,000
Materials handling	644,000	756,000
Inspection	804,000	396,000
Production costs	8,478,000	16,922,000
Units produced	100,000	200,000
	HK$	HK$
Production costs per unit	84.78	84.61
Mark-up (25%)	21.20	21.15
Selling price per unit	105.98	105.76

Answer 2(c)

Comparison on the selling prices:

	Premium	Deluxe
	HK$	HK$
Traditional absorption costing	94.50	111.50
Activity-based costing ("ABC")	105.98	105.76

Under ABC, the production costs of the two products are more or less the same and thus their selling prices, HK$105.98 for Premium and HK$105.76 for Deluxe.

Under traditional absorption costing, PPL earns much lesser revenue when it sells Premium at the price of HK$94.50. Premium is under-costed because it incurs higher levels of activity for set up and inspection under ABC, but does not absorb the corresponding higher overheads under traditional absorption costing.

On the other hand, the selling price for Deluxe of HK$111.50 may be too high (as a result of over-cost) to compete against competitors under traditional absorption costing. Deluxe should have absorbed corresponding lower overheads relating to set up and inspection because it consumes lower levels of activity for set up and inspection under ABC.

Answer 3(a)

Direct materials price variance:
AQ* × SP – AQ* × AP
= 59,000 × HK$20 – HK$1,298,000
= HK$118,000 Adverse
* at time of receipt

Direct materials quantity variance:
SP × (SQ – AQ#)
= HK$20 × (5,600 × 10 – 54,000)
= HK$40,000 Favourable
at time of usage

Answer 3(b)

Direct labour rate variance:
AH paid × SR – AH paid × AR
= 18,000 x HK$80 – HK$1,350,000
= HK$90,000 Favourable

Direct labour idle time variance:
SR × (AH worked – AH paid)
= HK$80 × (17,800 – 18,000)
= HK$16,000 Adverse

Direct labour efficiency variance:
SR × (SH – AH worked)
= HK$80 × (5,600 × 3 – 17,800)
= HK$80,000 Adverse

Answer 3(c)

Variable manufacturing overheads spending variance:
AH × SR – AH × AR
= (17,800 × HK$50) – HK$860,000
= HK$30,000 Favourable

Variable manufacturing overheads efficiency variance:
SR × (SH – AH)
= HK$50 × (5,600 × 3 – 17,800)
= HK$50,000 Adverse

Answer 3(d)

The benefit of the policy to calculate direct materials price variance at the time of receipt is that direct materials price variance is brought to the attention of the management much earlier. Moreover, all material inventories are then measured at standard cost that simplifies the valuation of inventories and recording of production costs.

Answer 3(e)

Labour production volume ratio:
(Production achieved in standard hours ÷ Budgeted standard hours) × 100%
= [(5,600 × 3 hours) ÷ 18,300 hours] × 100%
= 91.80%

Labour capacity ratio:
(Actual hours worked ÷ Budgeted standard hours) × 100%
= (17,800 hours ÷ 18,300 hours) × 100%
= 97.27%

Answer 4(a)

The simple objective of total quality management ("TQM") is "do the right things, right the first time, every time". TQM consists of organisation-wide efforts to strive to produce quality products and reduce expensive defects in the supply chain and to improve the value chain.

The four categories of cost of quality are prevention costs, appraisal costs, internal failure costs and external failure costs.

Answer 4(b)(i)

DIL is recommended to take both new projects as both residual incomes ("RIs") are positive. DIL as a whole will be better off if both new projects are taken despite that the return on investment ("ROI") of Shanghai Division's new project is lower than its current ROI of 15%.

Answer 4(b)(ii)

The manager of Guangdong Division will accept the new project as it generates a positive RI and the ROI is higher than its current ROI of 15%. The ROI of Guangdong Division after taking the new project must be over 15%.

However, the manager of Shanghai Division will reject the new project especially when his performance and / or remuneration is linked to the ROI. Since the ROI of the new project is below the current ROI of 15%, it would bring the division's overall ROI down to less than 15%, even though the project has a positive RI.

* * * END OF EXAMINATION PAPER * * *